# SC306 Homework 5

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## Question 1

**A)** $AR\ \% = \frac{R_e - R_u}{R_e} \times 100$

```
## [1] 77.1
```

The attributable risk percent of low birth weight from Cocaine use is 77.1%.

**B)** This means approximately 77% of the low birth weight risk for infants born to mothers who used Cocaine while pregnant is attributable to their use of Cocaine.

**C)** $PAR\ \% = \frac{P_e(R_e - R_u)}{P_e R_e + (1 - P_e) R_u} \times 100$

```
## [1] 6.3
```

The population attributable risk percent of low birth weight from Cocaine use is 6.3%.

**D)** Approximately 6.3% of the low birth weight risk for all infants born in California is attributable to the use of Cocaine while pregnant.

# Code Appendix

```r
knitr::opts_chunk$set(
  echo = FALSE,                 # don't show code
  warning = FALSE,              # don't show warnings
  message = FALSE,              # don't show messages (less serious warnings)
  cache = FALSE,                # set to TRUE to save results from last compilation
  fig.align = "center",         # center figures
  attr.source = '.numberLines'  # add line numbers to code
)
library(tidyverse)
library(dplyr)
library(knitr)
library(epitools)

set.seed(24)

# Question 1
# Part A
Re = (110)/(110+289) # Risk among exposed

Ru = (1073)/(1073+15896) # Risk among unexposed

AR = (Re - Ru)/(Re) * 100 # Attributable risk

round(AR, 1)
# Part C
Pe = 0.02
PAR = (Pe * (Re - Ru)) / (Pe * Re + (1-Pe) * Ru) * 100
round(PAR, 2)
```

Attributions: I have used some template/styling code from GitHub user alexpghayes.